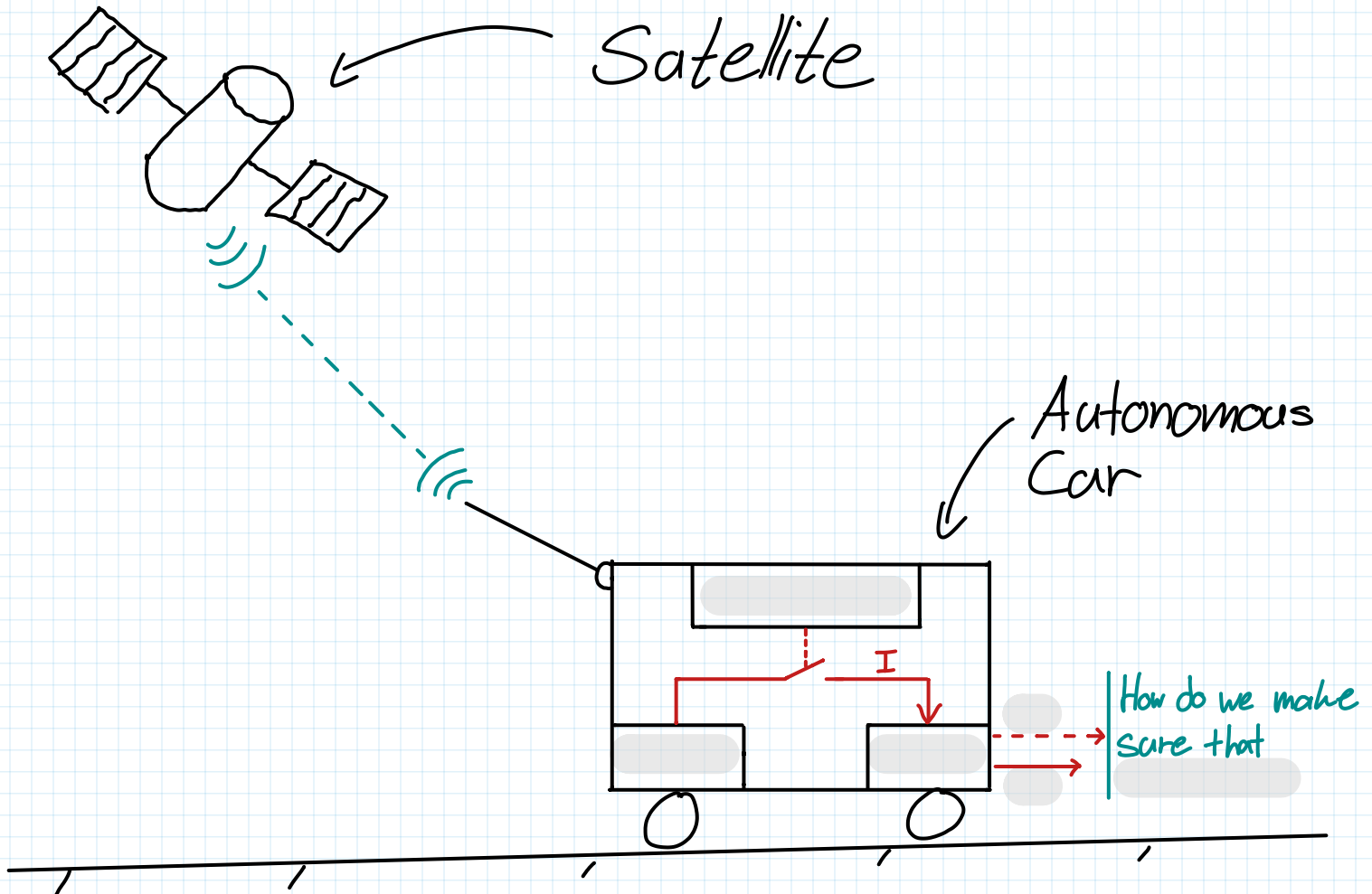


What is control?

- Let's build intuition with an example



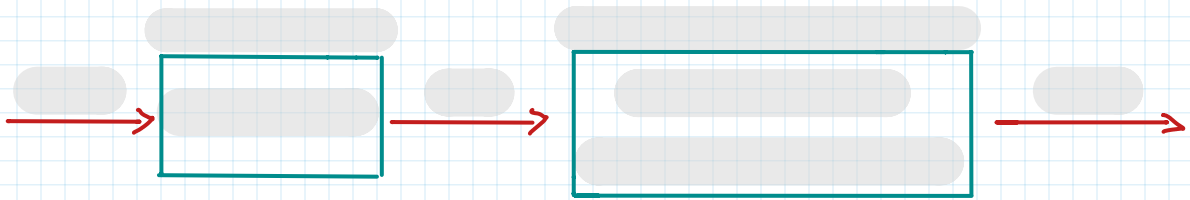
- We command the car's computer to move the car with .
- The computer can only control the going into the motor.
- The computer knows the car's true speed through of the satellite.

How does the computer know how much current it should send?

⇒ This is the central question of control

• To solve this we will first need to learn many techniques. For you to keep an overview, you must keep the greater picture in mind.

⇒ Let's abstract the car example
(For now without the satellite)



• V_o , I , V_m are called _____.

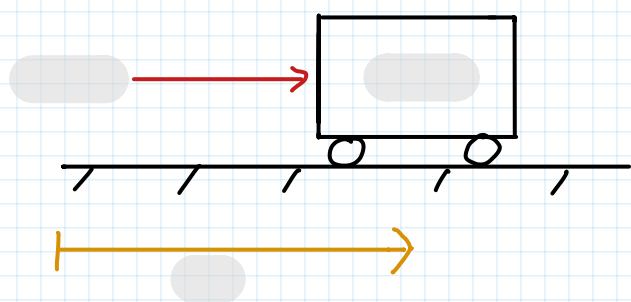
• Computer, Motor are called _____.

⇒ A system _____ a signal

What are signals and systems?

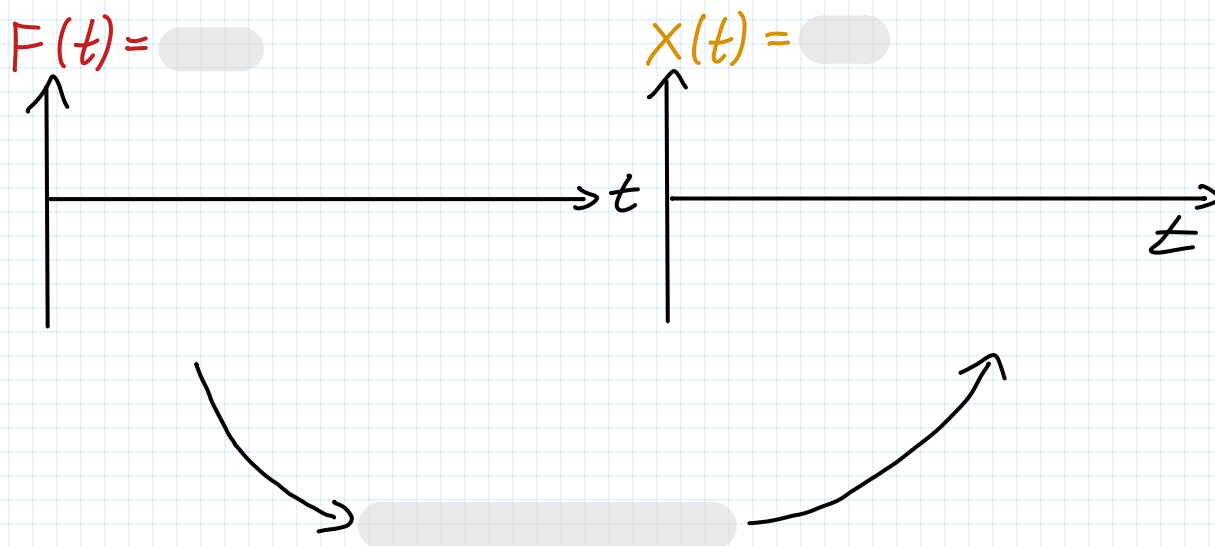
- Signal: A _____ of _____ (or _____)
- System: Any _____ that possibly has _____ and/or _____

Example:



- Mass m
- force $F(t)$
- position x

- System: The _____ with mass m
- Signals: _____, _____



- We can model systems with differential equations

Here:

$\Rightarrow$

$$\Rightarrow \dot{x}(t) =$$

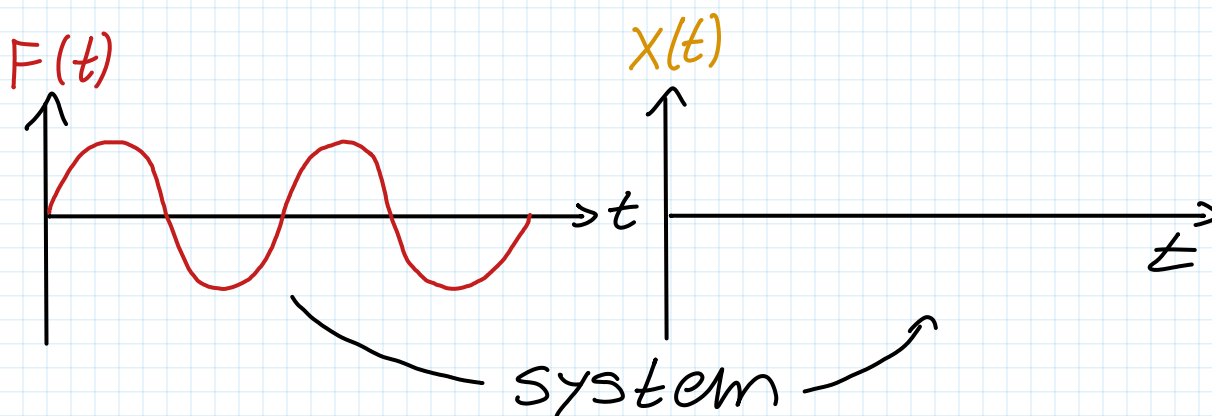
$$\Rightarrow x(t) =$$

Assume
 $x_0 =$, $v_0 =$

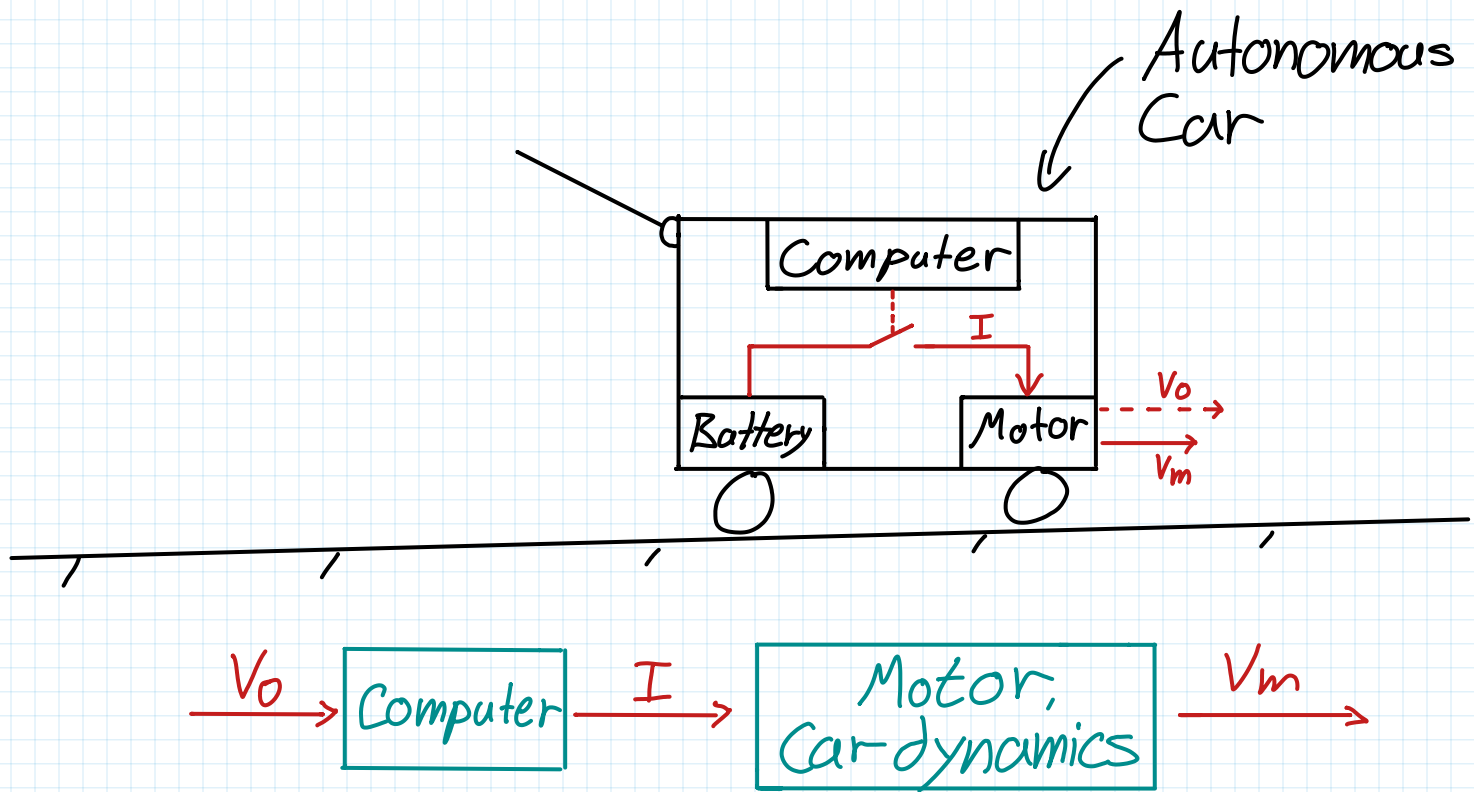
From initial conditions: $C_1 =$, $C_2 =$

$$x(t) =$$

$\Rightarrow$ The system is the underlying equation and the . It defines how the input signal is manipulated.



Wait, why did we just look at this?



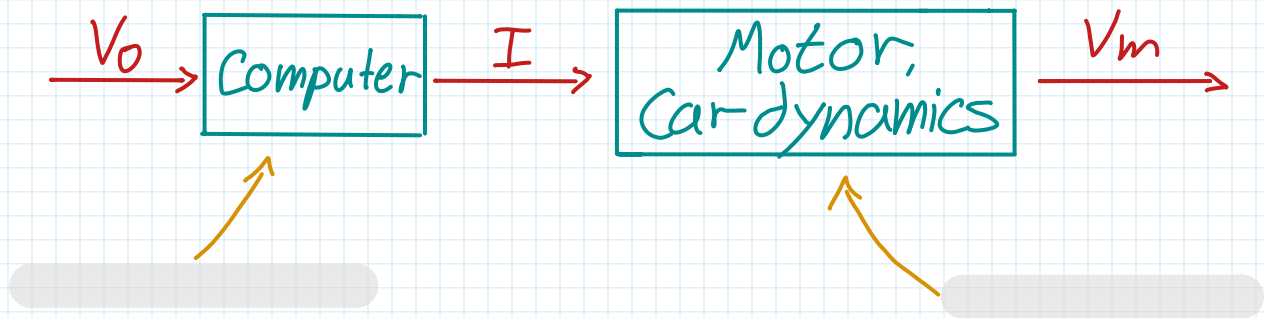
- : Blocks are systems
- →: Arrows are signals

⇒ To figure out how to control the car, we need to know the diff' eqns that describe the behaviour of the blocks!

- This is why the first part of the course will be mathematical.

If every system is just defined by physics, then what even is our job as engineers?

⇒ This is not true!!



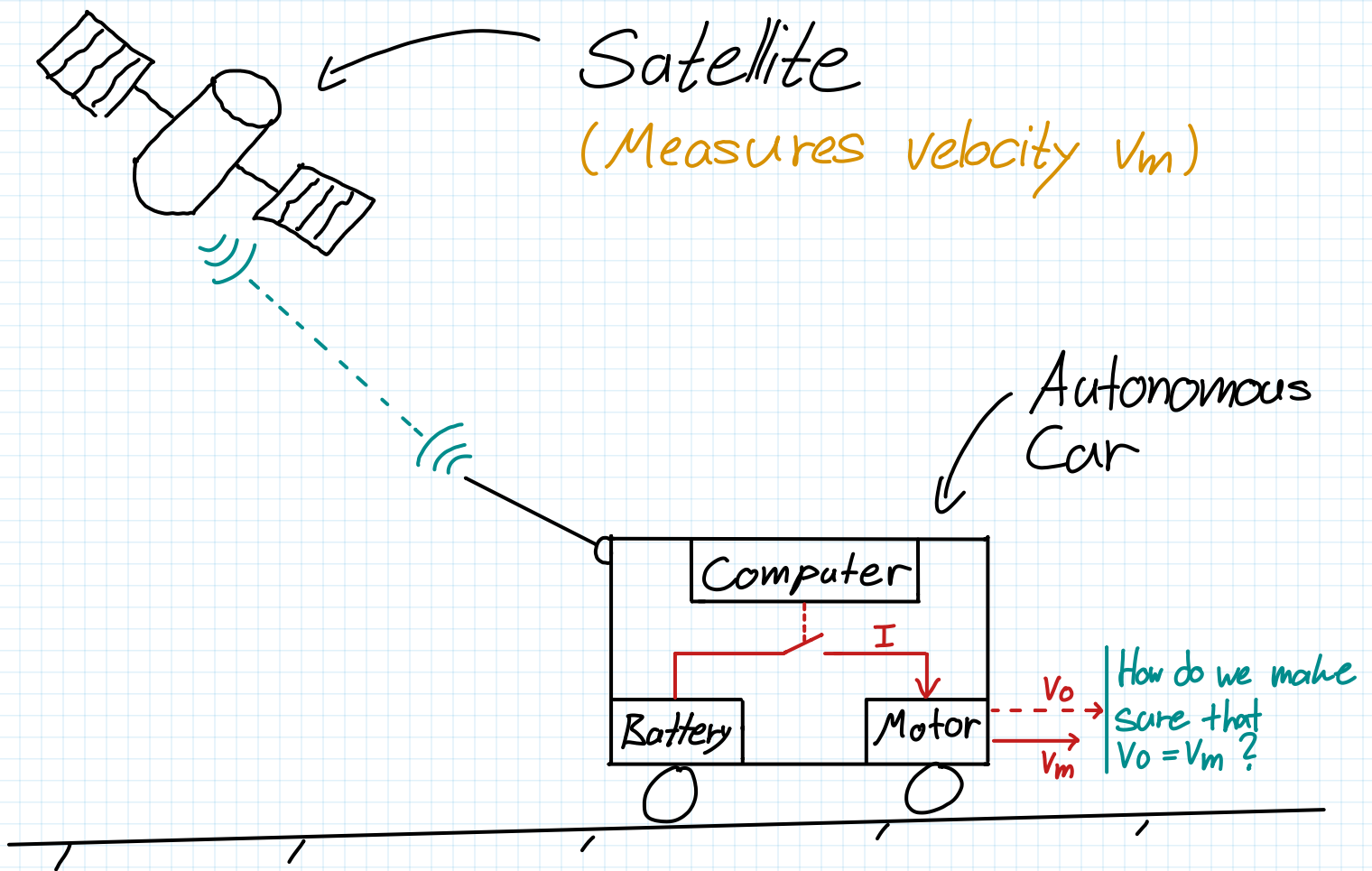
• Some blocks are predefined. The ones that we can design (*computer*), are called .

⇒ We need to design controllers such that the does what we want. This is hard because of:

-
-
-
- ...

• These problems will be addressed by introducing .

⇒ Back to our example



Think about how the satellite could improve this problem...

- All of what I showed you was massively simplified and not covered yet.
(you are not expected to understand it now)
- ⇒ This was just so you have the big picture in mind!

Intermediate Quiz

Q1) The Input is what we tell the computer to do.

☐ True ☐ False

Q2) Each block represents a physical system. ☐ True ☐ False

Q3) Not all control systems are electrical.

☐ True ☐ False